



Google

DiffusionGemma: Google's open model that rewrites how text is generated

[swipe →](#)



1

Not word by word

DiffusionGemma starts with a canvas of noise and refines 256 tokens simultaneously. Every standard model goes one word at a time. This one doesn't.



2

The speed is real

1,000 tokens/sec on an H100. 700+ on an RTX 5090. About 4x faster than autoregressive Gemma 4. Fits 18GB VRAM when quantized. Apache 2.0.



3

The quality gap is also real

Scores lower on every published benchmark —
MMLU, GPQA Diamond, LiveCodeBench v6.

Google's own docs say: use standard Gemma 4
when quality matters most.



4

Where it actually fits

Code infilling, text insertion, structured data tasks.
Speed-critical jobs where some quality loss is
tolerable. Not your main reasoning model — yet.



The real question

Image diffusion models kept getting better until they beat everything. Does text diffusion follow the same curve?

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